Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(Empowered Autonomous Institution under R.T.M Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++ Grade
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu
Department of CSE-IOT

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering: Dynamic learning environment that encourages problem-solving using emerging technology, · National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration, Holistic, ethical and sustainable development for lifelong success.

Session 2025-2026(EVEN)

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

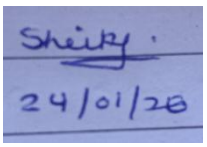
Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.



Sheikh Wasimuddin
Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	**2025-26 (EVEN)**	**Course Name**	**Mobile and Web Application for IOT Lab**
Semester	**6**	**Course Code**	**23IOT1624**
Roll No	**63**	**Name of Student**	**Sheikh Wasimuddin**

Practical Number	**03**
Course Outcome	1. Understand and apply basic Kotlin programming constructs including variables, control structures, functions, null safety, and object-oriented concepts for Android app development. 2. Design and develop responsive user interfaces using XML layout files, View Groups, and widgets in Android Studio, adhering to Material Design principles. 3. Implement event handling and application logic using Kotlin to capture user input, manage app state, navigate between screens, and interact with Android system components like Toasts and Dialogs 4. Integrate learned concepts to develop a mini Android application that demonstrates use of ViewModel, multi-screen navigation, and proper UI/UX design for real-world problem-solving
Aim	Student Result Analyzer
Problem Definition	You are given a list of student names with their marks. Perform the following tasks using Kotlin collections and higher-order functions: 1. Store student names and marks using a Map. 2. Display all students using forEach. 3. Filter students who scored 60 or above. 4. Add 5 grace marks to the filtered students. 5. Create a custom higher-order function that accepts marks and a grading logic. 6. Assign grades: ○ A → marks ≥ 85 ○ B → marks ≥ 70 ○ C → marks ≥ 60 ○ Fail → below 60

Theory (100 words)	Kotlin offers a flexible collection framework that simplifies data storage and manipulation. The most commonly used collections are List, Set, and Map. A List stores elements in an ordered manner and allows duplicate values, making it useful when the sequence of data matters, such as storing student marks. Lists are created using listOf() and traversed using forEach() with lambda expressions. A Set stores only unique elements and automatically removes duplicates, which is helpful in cases like storing roll numbers or unique identifiers. Sets are created using setOf() and accessed through forEach(). A Map stores data as key–value pairs, where each key is unique and maps to a specific value, such as mapping names to marks. Maps are created using mapOf() and accessed using destructuring in loops. Kotlin also supports lambdas and higher-order functions. Lambdas are anonymous functions, while higher-order functions accept functions as parameters. Functions like filter() and map() enable efficient and readable data processing.
Procedure and Execution (100 Words)	Step for Implementation: 1. Begin by creating a Kotlin/JVM project and write the `main()` function, which serves as the starting point of program execution. 2. Use a Map collection to store student information, where student names are used as keys and their marks are stored as corresponding values. 3. Display all the student details by traversing the map using the `forEach` function. 4. Apply the `filter()` higher-order function to select only those students who have obtained 60 marks or more. 5. Increase the marks of the filtered students by adding 5 grace marks using the `mapValues()` function. 6. Define a custom higher-order function that takes student marks and a lambda expression representing grading logic as parameters. 7. Inside the lambda expression, assign grades such as A, B, C, or Fail based on the updated marks using conditional statements. 8. Finally, display the output showing each student's name, revised marks, and assigned grade in a clear format.

Code:

```
//sheikhwasimuddin 23071044 CSEIOT_63
fun main() {
   val students = mapOf(
      "sheikh" to 75,
      "Tejas" to 60,
      "Tanmay" to 91,
      "Karan" to 40,
      "parth" to 20
   )
   println("1. All Students and Marks:")
   students.forEach { (name, marks) ->
      println("$name -> $marks")
   }
   val passedStudents = students.filter { it.value >= 60 }
   println("\n2. Students Scoring 60 or Above:")
   passedStudents.forEach { (name, marks) ->
      println("$name -> $marks")
   }
   val graceMarks = passedStudents.mapValues { it.value + 5 }
   println("\n3. After Adding 5 Grace Marks:")
   graceMarks.forEach { (name, marks) ->
      println("$name -> $marks")
   }
   fun calculateGrade(marks: Int, gradeLogic: (Int) -> String):
String {
      return gradeLogic(marks)
   }
   println("\n4. Final Result with Grades:")
   students.forEach { (name, originalMarks) ->
      val finalMarks = if (originalMarks >= 60) originalMarks +
5 else originalMarks
      val grade = calculateGrade(finalMarks) {
         when {
            it >= 85 -> "A"
            it >= 70 -> "B"
            it >= 60 -> "C"
            else -> "Fail"
         }
      }
      println("$name -> Marks: $finalMarks -> Grade: $grade")
   }}
```

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Output:

```
//sheikhwasimuddin 23071044 CSEIOT_63
fun main() {

    val students = mapOf(
        "sheikh" to 75,
        "Tejas" to 60,
        "Tanmay" to 91,
        "Karan" to 40,
        "parth" to 20
    )

    println("1. All Students and Marks:")
```

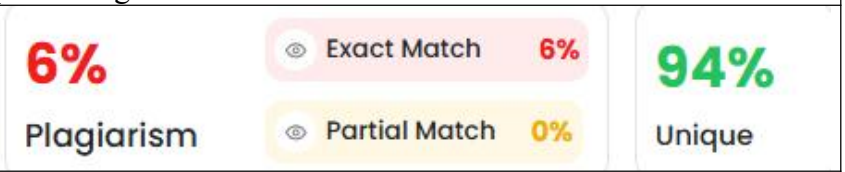
```
1. All Students and Marks:
sheikh -> 75
Tejas -> 60
Tanmay -> 91
Karan -> 40
parth -> 20
```

```
//sheikhwasimuddin 23071044 CSEIOT_63
fun main() {

    val students = mapOf(
        "sheikh" to 75,
        "Tejas" to 60,
        "Tanmay" to 91,
        "Karan" to 40,
        "parth" to 20
    )

    println("1. All Students and Marks:")
```

```
2. Students Scoring 60 or Above:
sheikh -> 75
Tejas -> 60
Tanmay -> 91

3. After Adding 5 Grace Marks:
sheikh -> 80
Tejas -> 65
Tanmay -> 96
```

	``fun main() {
	//sheikhwasimuddin 23071044 CSEIOT_63
	val students = mapOf(
	"sheikh" to 75,
	"Tejas" to 60,
	"Tanmay" to 91,
	"Karan" to 40,
	"parth" to 20
	)
	println("1. All Students and Marks:"
	sheikh -> 80
	Tejas -> 65
	Tanmay -> 96
	4. Final Result with Grades:
	sheikh -> Marks: 80 -> Grade: B
	Tejas -> Marks: 65 -> Grade: C
	Tanmay -> Marks: 96 -> Grade: A
	Karan -> Marks: 40 -> Grade: Fail
	parth -> Marks: 20 -> Grade: Fail```
	Output Analysis

Output Analysis	The program begins by creating a Map that stores student names as keys and their corresponding marks as values. In the first step, all student records are displayed using the forEach function. This output shows the original marks of each student, including those who scored below the passing criteria.In the second step, the filter() higher-order function is applied to select only those students who have scored 60 marks or above**.** As a result, students with marks below 60 are excluded from this list. The filtered output displays only eligible students.Next, the program adds 5 grace marks to the filtered students using the mapValues() function. This updated list shows increased marks only for students who passed the filter condition, while others remain unchanged.Finally, a custom higher-order function is used to assign grades based on the updated marks. Conditional logic inside a lambda expression assigns grades A, B, C, or Fail. The final output displays each student's name along with their final marks and assigned grade, clearly indicating overall performance.
Link of student Github profile where lab assignment has been uploaded	**https://github.com/sheikhwasimuddin**

Conclusion	This program successfully demonstrates the effective use of Kotlin collections and higher-order functions to manage and process student data. By using a Map, student names and marks are stored in an organized key–value format. The forEach function enables simple and readable traversal of records, while the filter() function efficiently selects students who meet the passing criteria. The use of mapValues() allows graceful modification of marks by adding grace points only to eligible students. Additionally, the implementation of a custom higher-order function with lambda expressions showcases Kotlin's functional programming capabilities by dynamically applying grading logic. Overall, the program highlights how Kotlin's collection framework and higher-order functions help in writing clean, modular, reusable, and efficient code for real-world data processing tasks.
Plag Report (Similarity index < 12%)	
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Observation copy screenshot

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Practical No :- 03

Aim: Student Result Analyzer

Problem statements: you are given a list of students names with their marks. Perform following tasks using kotlin collections and higher order functions:

1. store student name & marks using Map.
2. Display all students using forEach.
3. filter who scored 60 or above.
4. Add 5 grace marks to filter students
5. Create a custom higher order function
6. Assign grades:
   - A → marks ≥ 85
   - B → marks ≥ 70
   - C → marks ≥ 60
   - fail → marks below 60.

Theory :-

1. list: store elements in an ordered manner and allow duplicate value.
2. set: stores unique elements, auto removes duplicate.
3. Map: Store data in key value pair.
4. foreach: Iterates through each element of collections
5. map(): Transform each element of a collection

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(6) Lambda function: Anonymous function used for short logic

(7) higher order function: An function that Accept another function as a parameter.

output:-

(1) All students and Marks:
Sheikh → 75
Tejas → 60
Tanmay → 92
Karan → 40
Parth → 20

(2) students Scoring 60 or above:
Sheikh → 75
Tejas → 60
Tanmay → 92

(3) After Adding 5 grace Marks:
Sheikh → 80
Tejas → 65
Tanmay → 97

(4) final Result with Grades:
Sheikh → Marks: 80 → Grade: B
Tejas → Marks: 65 → Grade: C
Tanmay → Marks: 97 → Grade: A

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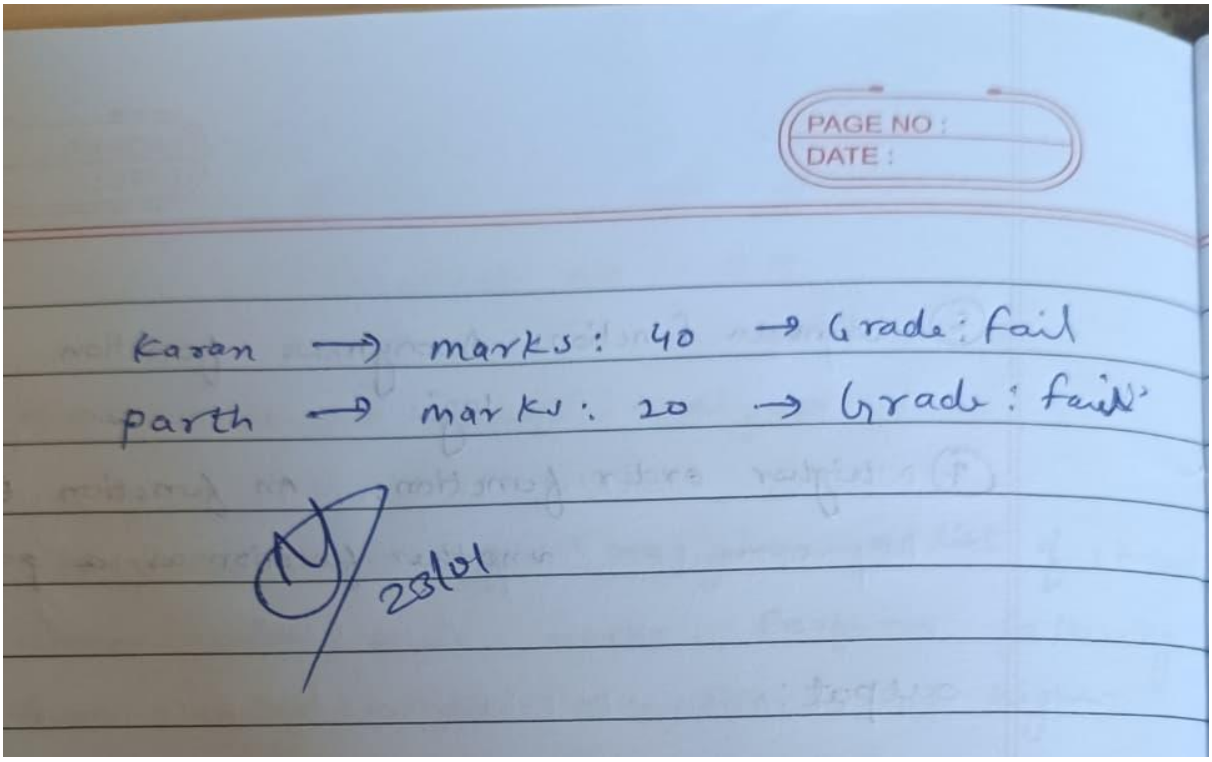
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